

FaultGuard AI - User Documentation

1. Overview

FaultGuard AI is an intelligent system designed to detect, classify, and locate electrical faults in power transmission lines using 1D Convolutional Neural Networks (1D-CNN).

2. Input Requirements

The system expects a CSV file with exactly 7 columns:

- Time (s)
- Va, Vb, Vc (Voltage p.u.)
- Ia, Ib, Ic (Current p.u.)

Sampling Frequency: 10 kHz (dt = 0.0001s).

3. Operational Workflow

1. Navigate to the Lab section.
2. Upload your CSV file or use the calibrated "Normal Data" and "Fault Data" buttons.
3. The AI model will process a 200-sample window (20ms) to provide diagnostics.
4. View results: Verdict, Type, Location (0-5 KM), and Inception Time (MS).

4. Detection Thresholds

Fault inception is triggered when:

- Voltage Peak < 0.85 p.u.
- OR Current Peak > 0.3 p.u.

The normal operating current peak is calibrated at 0.16 p.u.